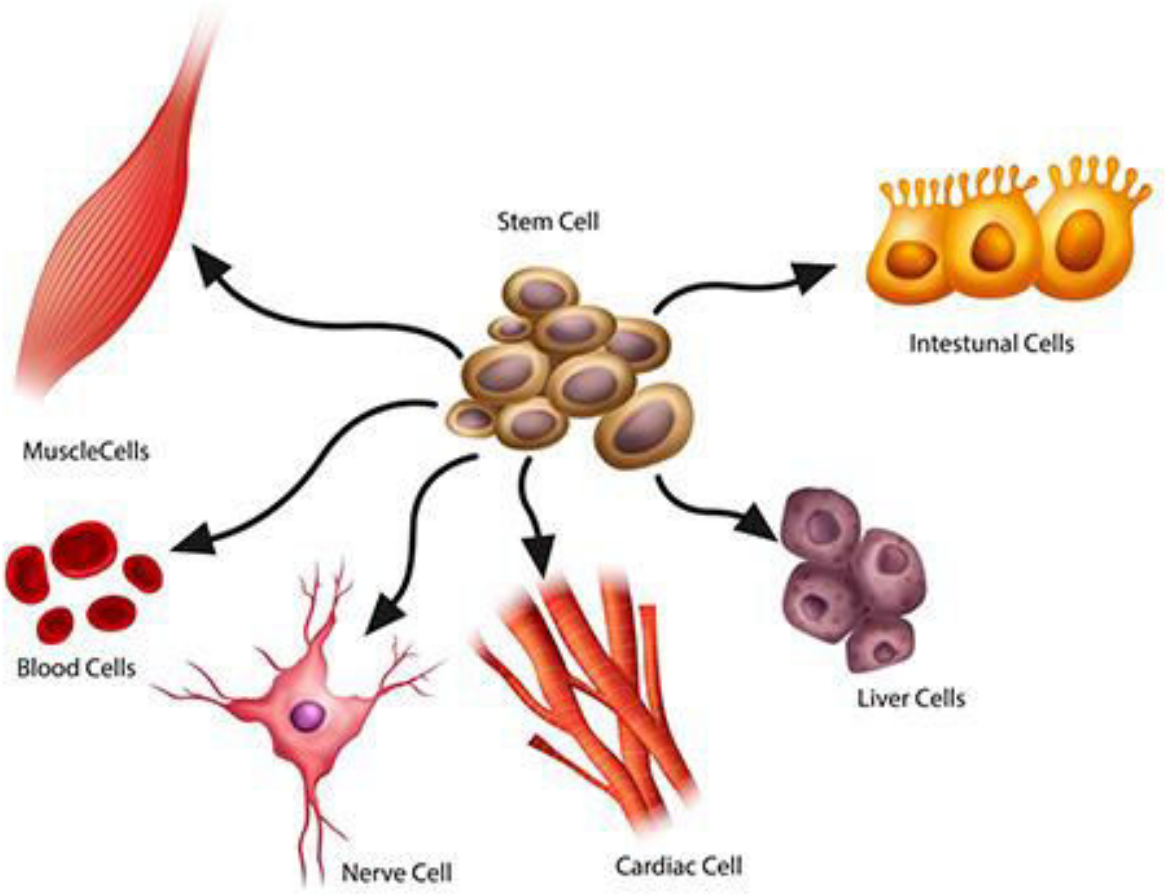
Chapter VII:
Introduction to Epigenetics

Differentiation



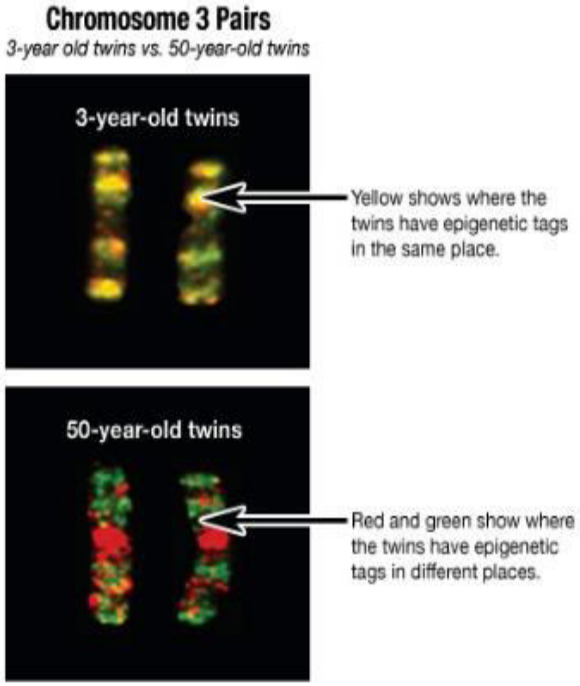
Q: Almost all our cells have the same DNA code. Why are they so different?

Growing up of identical twins



Environmetal factor

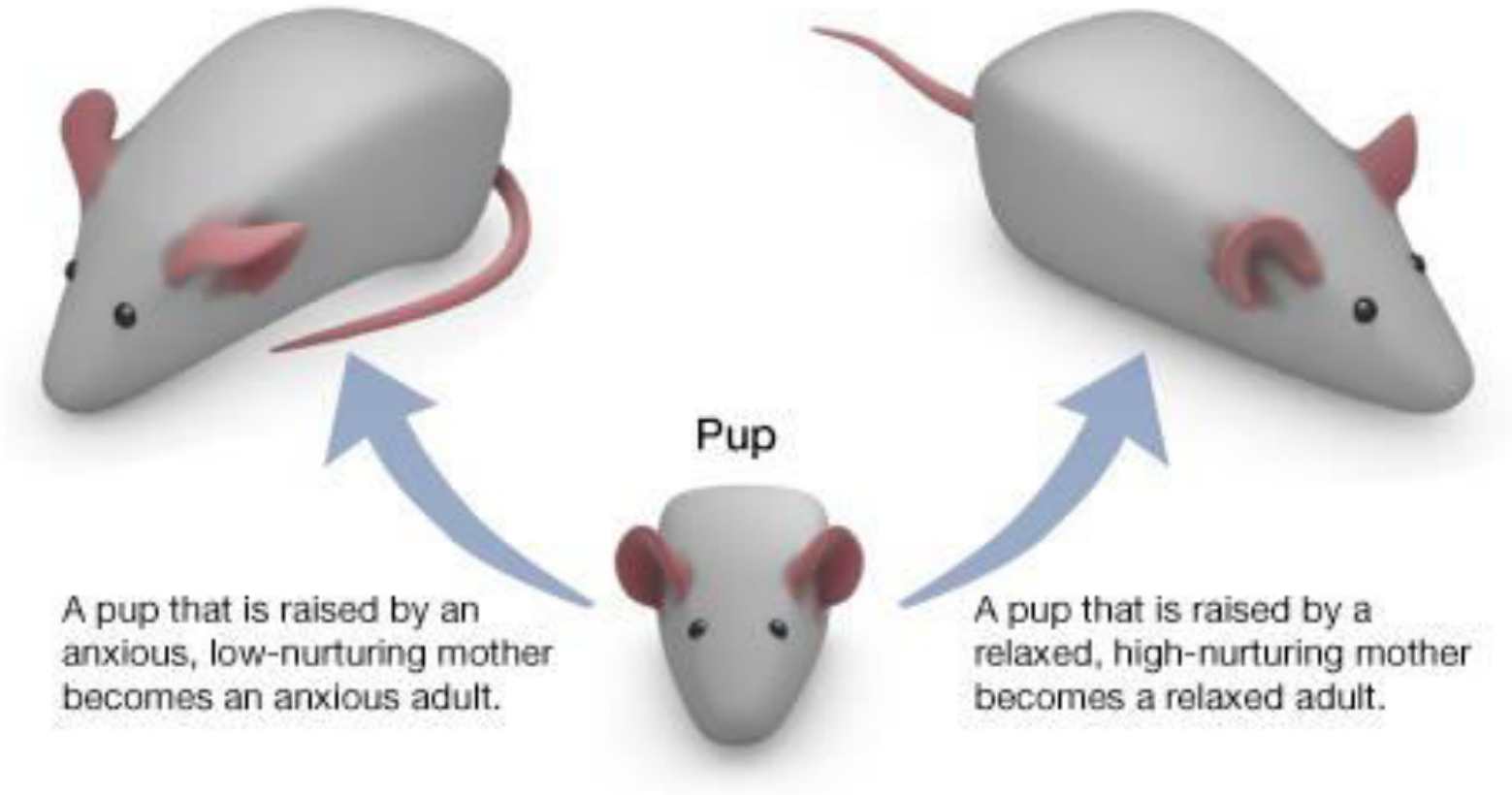




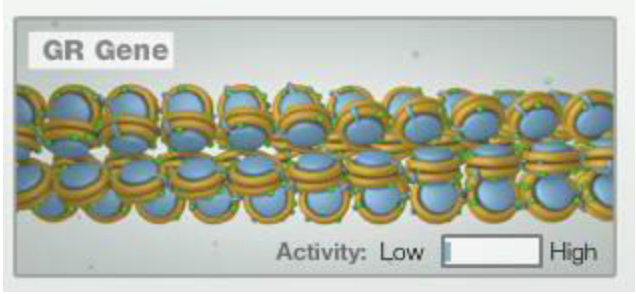
Chromosome 3 pairs in each set of twins are digitally superimposed. One twin's epigenetic tags are dyed red and the other twin's tags are dyed green. When red and green overlap, that region shows up as yellow. The 50-year old twins have more epigenetic tags in different places than do 3-year-old twins.

Q: These two siblings are identical twins (have exact same DNA). Why they're look different when they grow up?

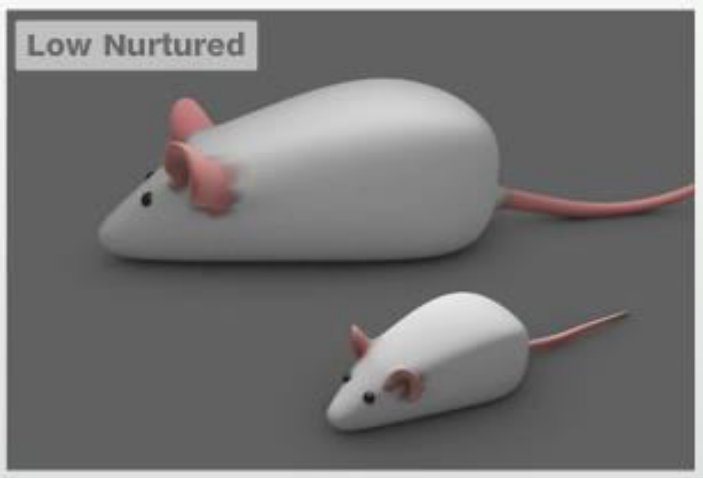
An example of epigenetic

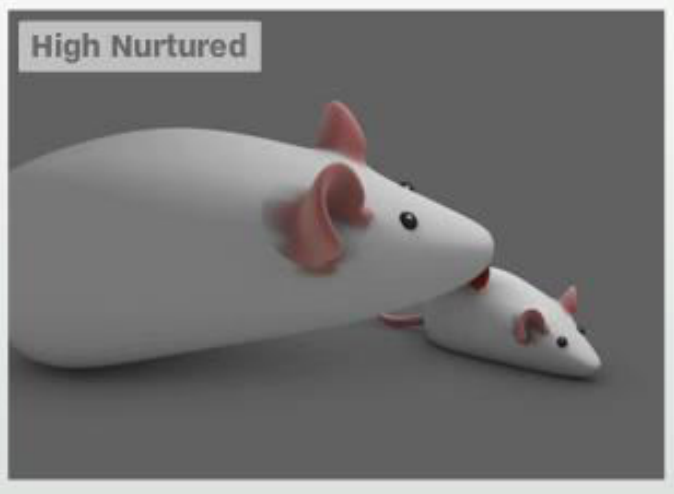


An example of epigenetic









ANXIOUS, STRESSFULL MOUSE

RELAXED, CALM MOUSE

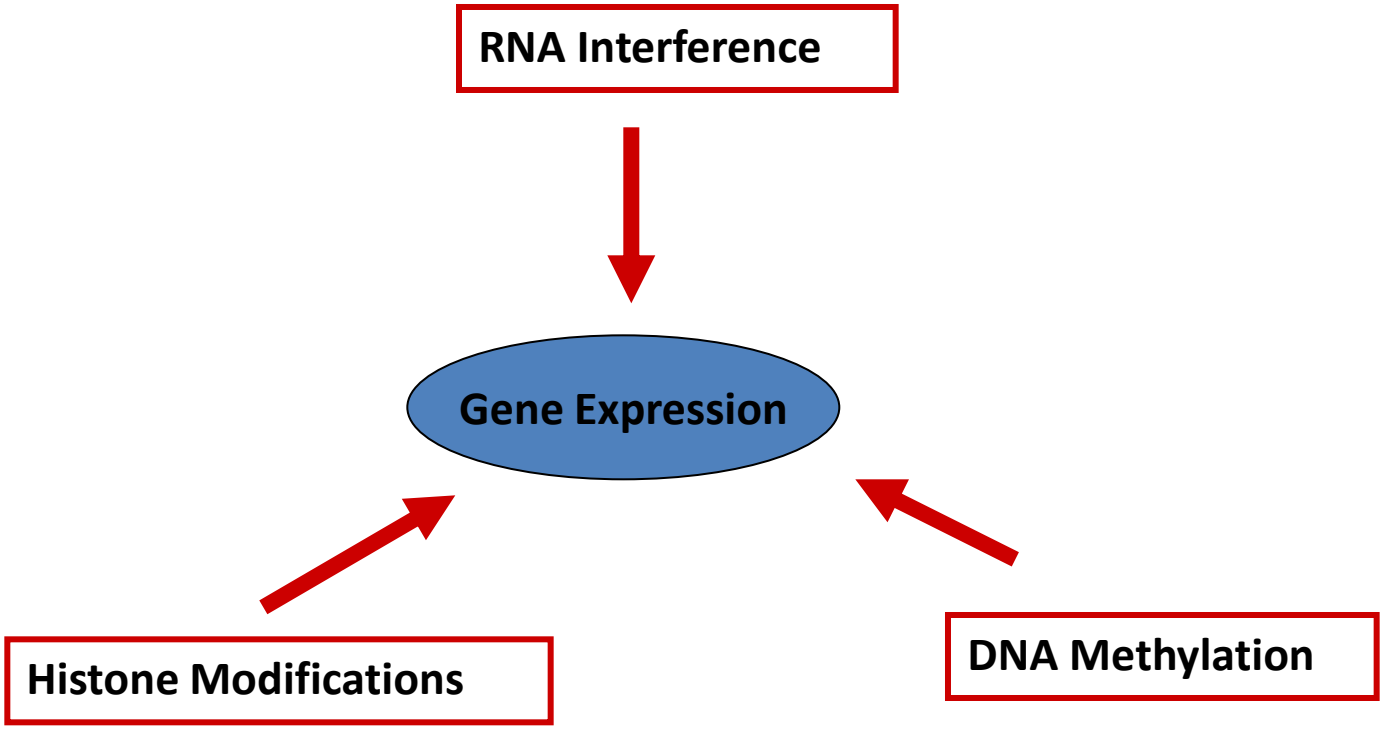
Epigenetics

- Epigenetics refers to the study of changes in the regulation of gene activity and expression that are not dependent on gene DNA sequence.

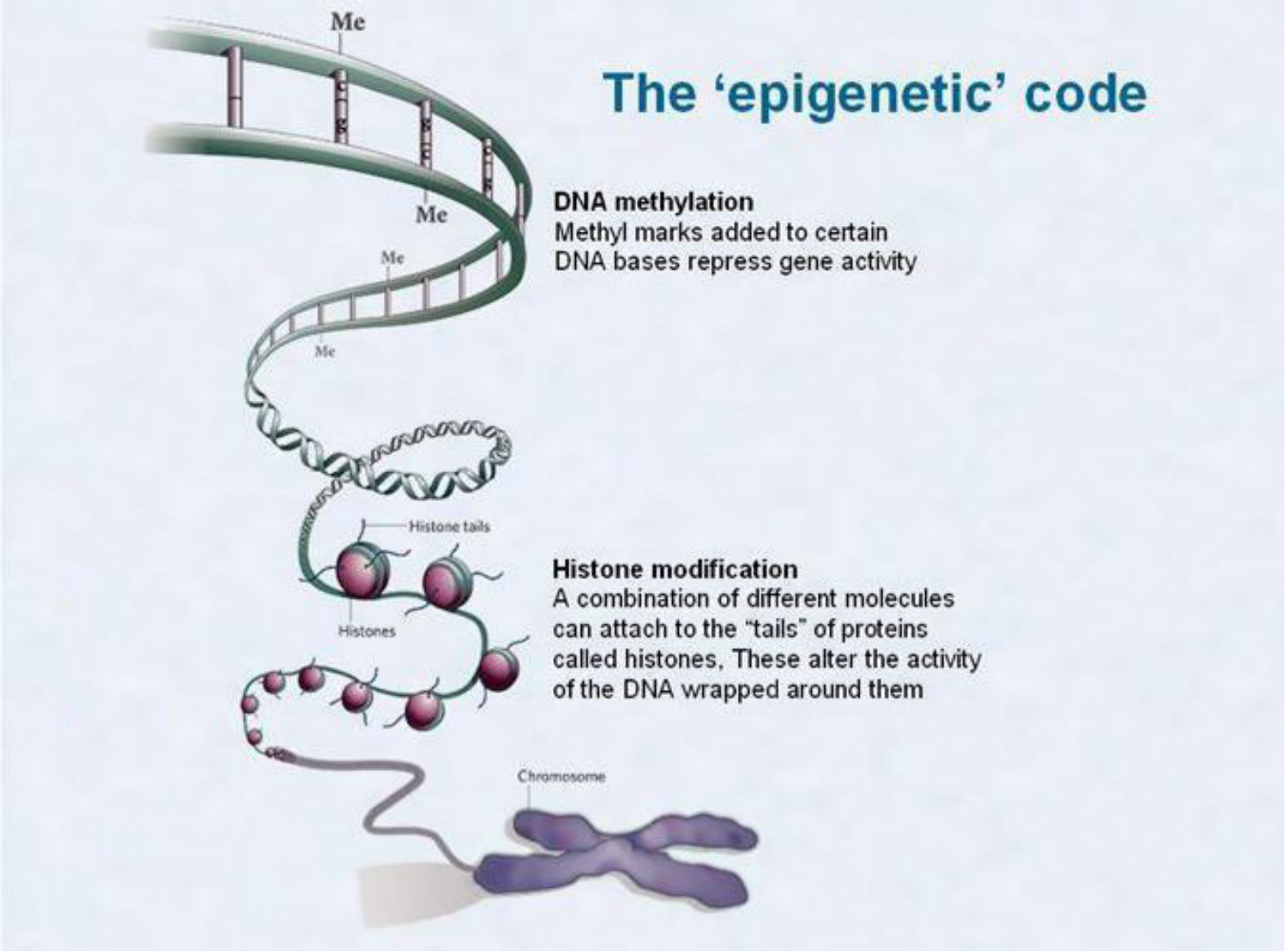
- While epigenetics often refers to the study of single genes or sets of genes, e**pigenomics** refers to more global analyses of epigenetic changes across the entire **genome.**

Epigenetics

03 basic mechanisms of epigenetics:

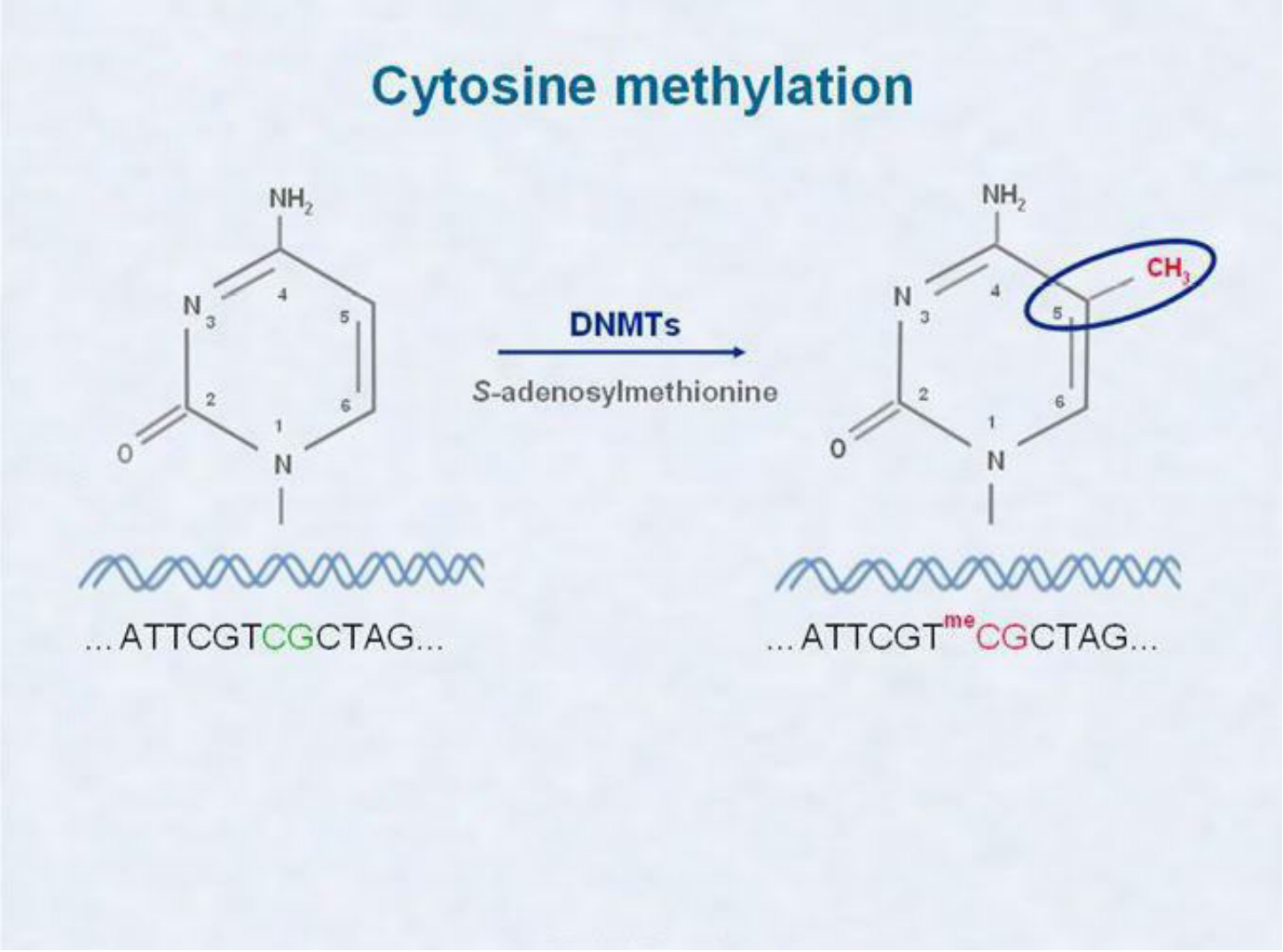


DNA Methylation & Histone Modifications Form the Epigenetic Code

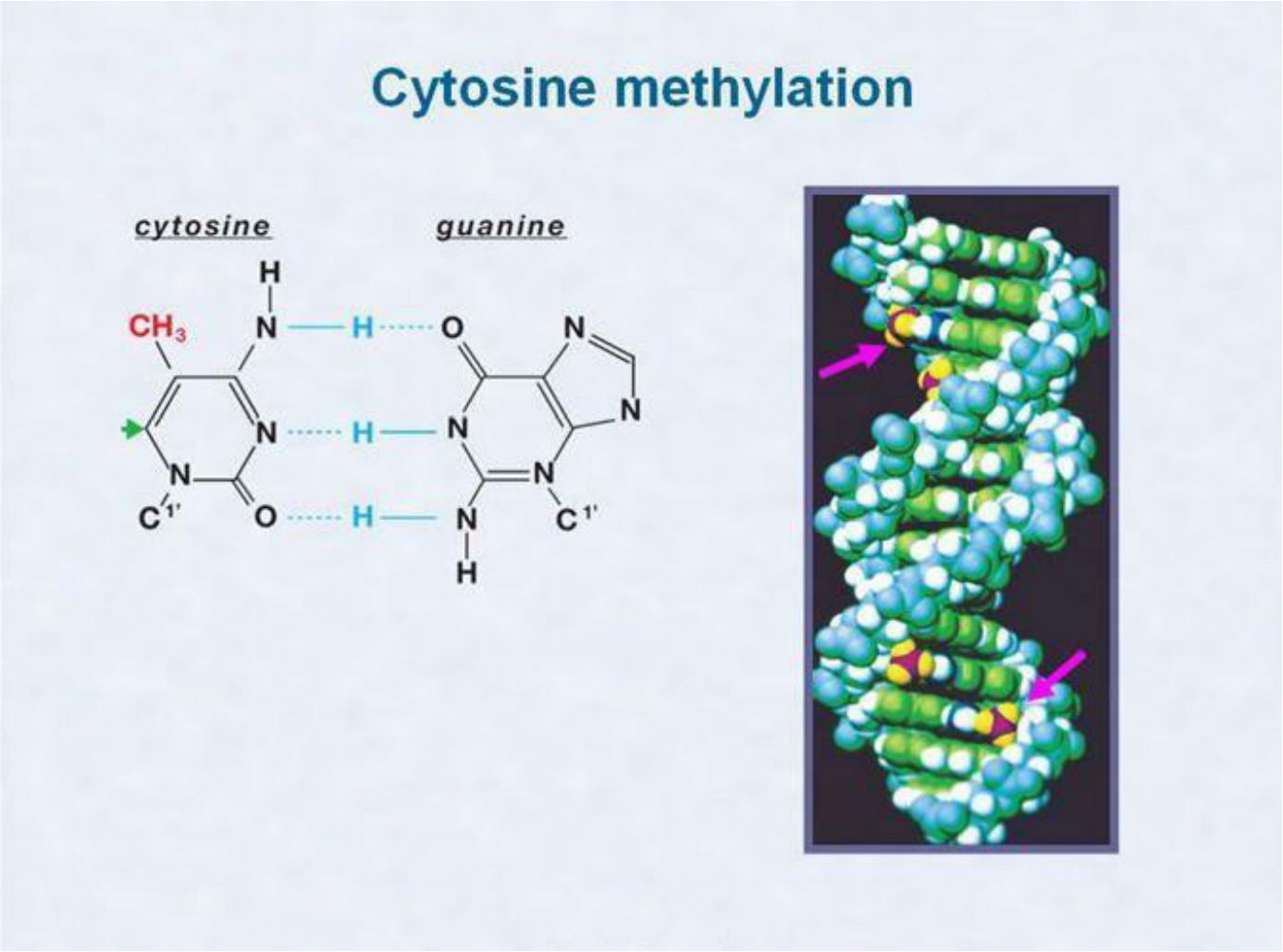


DNA methylation

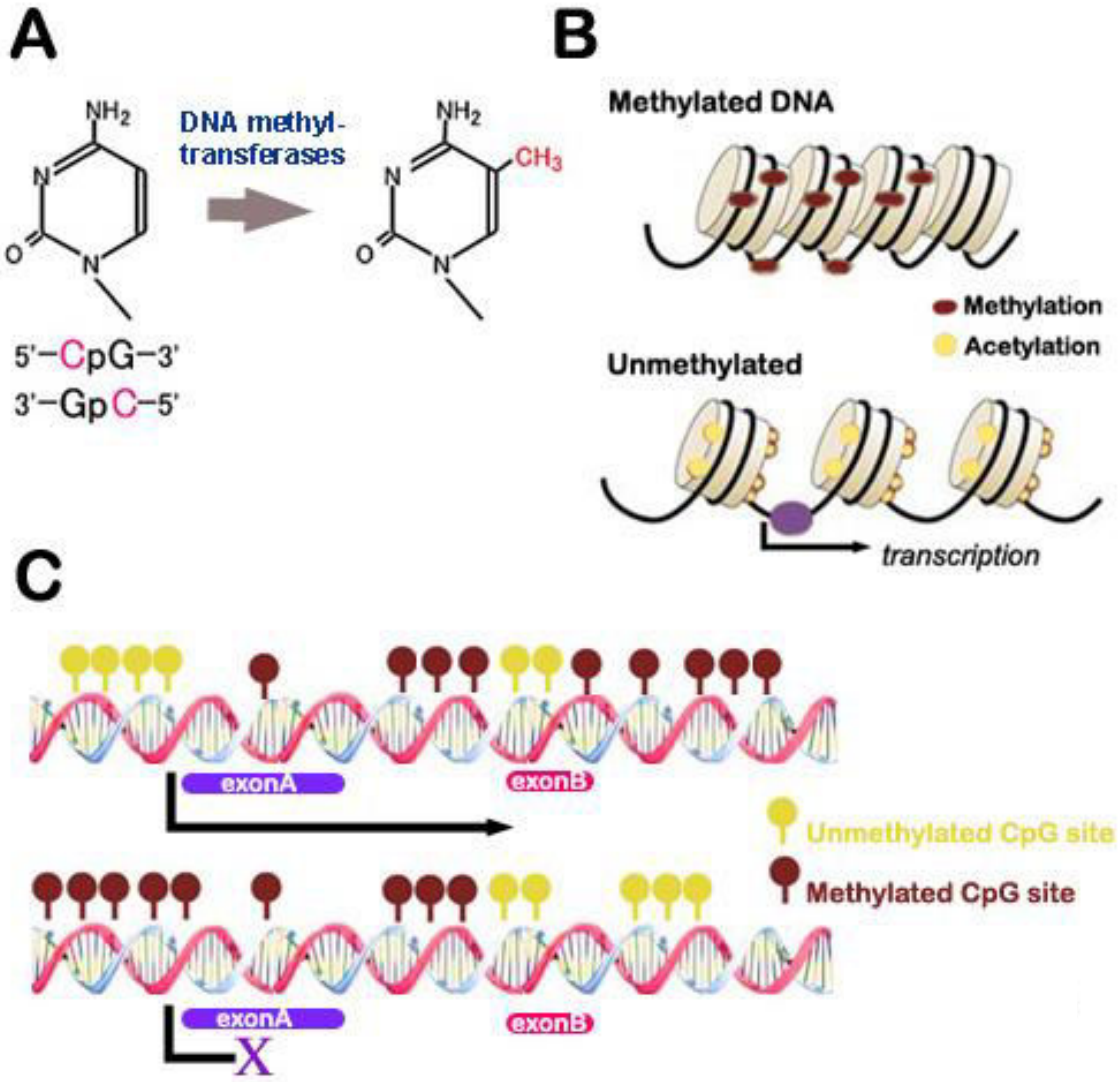
5-Methylation Cytosine in DNA



5-Methylation Cytosine in DNA



DNA Methylation



Hypomethylation
Hypermethylation

CpG island

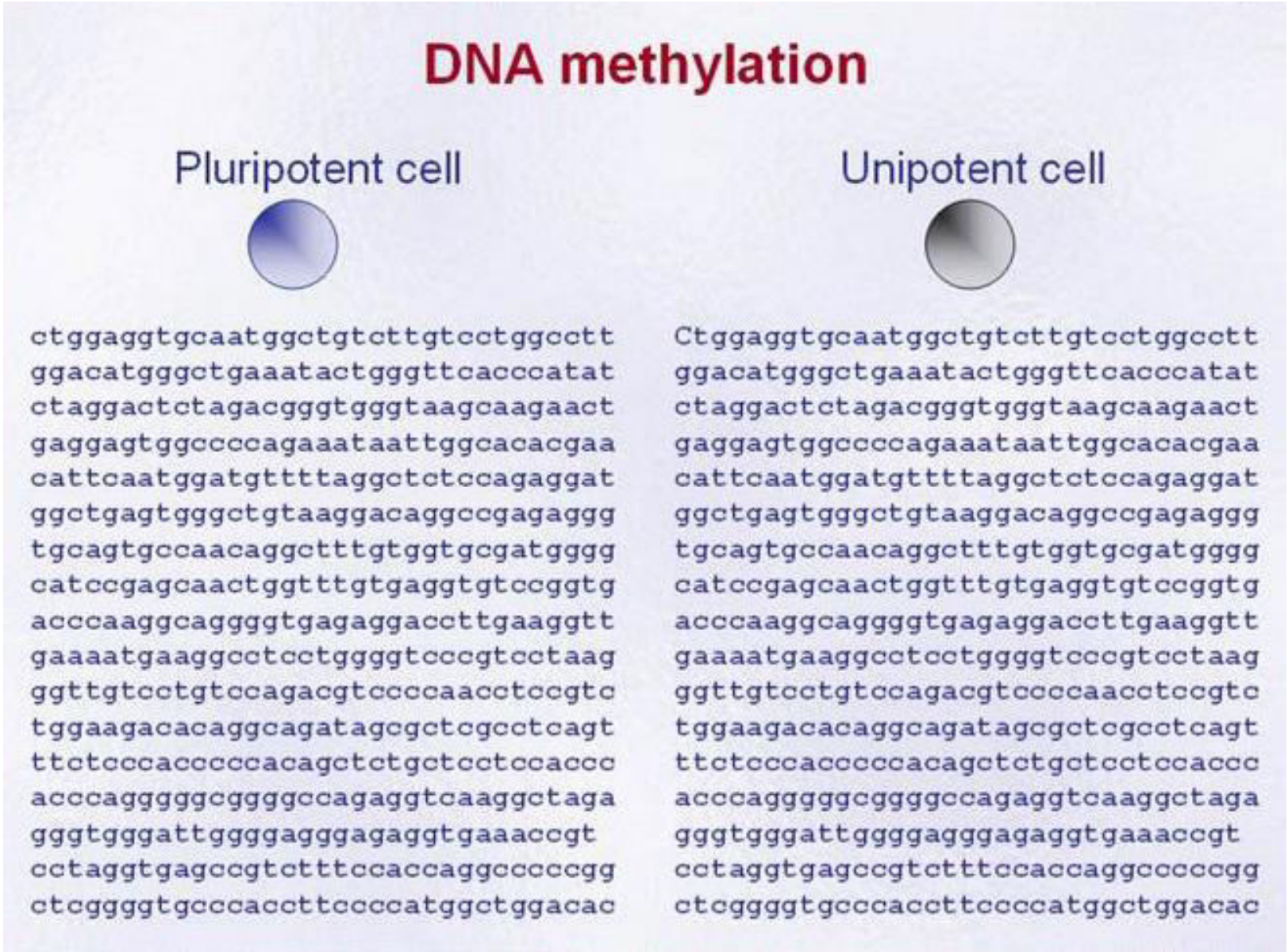
- Regions of the genome with high frequency of CpG sites than the rest of the genome.
- CpG island located in the promoter region of genes can play an important roles in gene silencing.
- In mammals, 70% to 80% of CpG cytosines are methylated



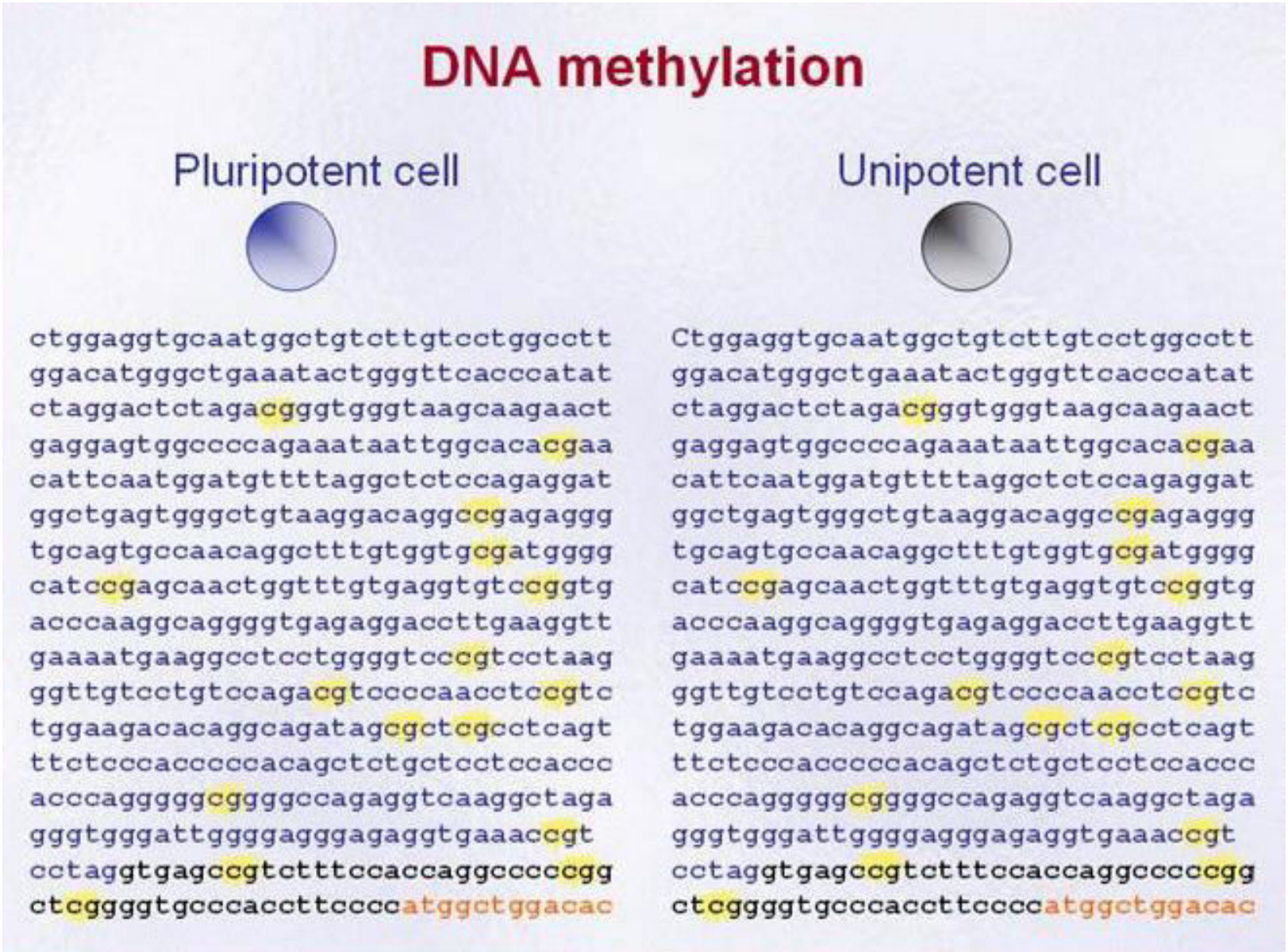
Natural Roles of DNA Methylation in Mammalian System

- Imprinting
- X chromosome inactivation
- Heterochromatin maintenance
- Developmental controls
- Tissue specific expression controls

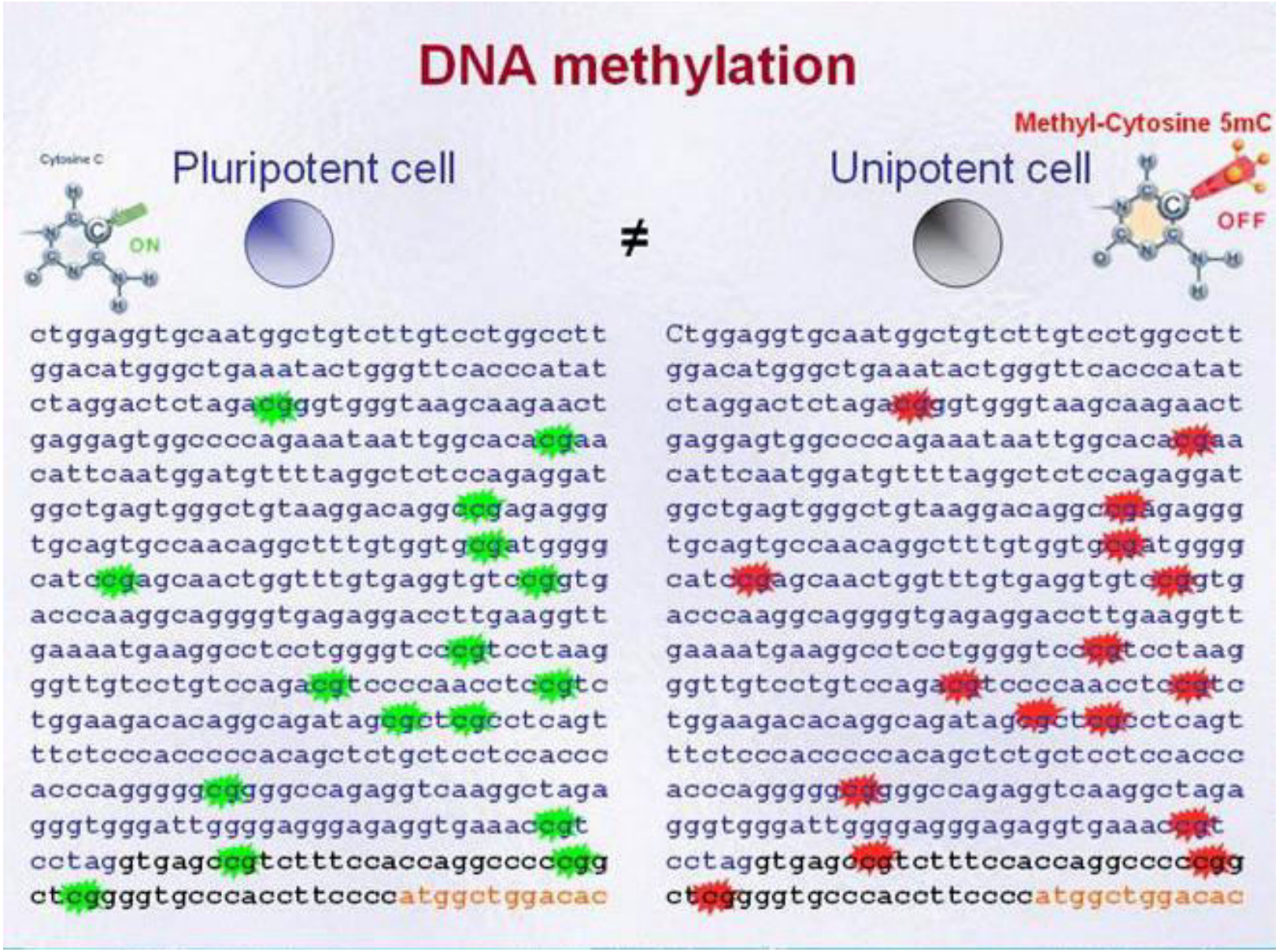
DNA Methylation Differentiates Totipotent Embryonic Stem Cells from Unipotent Adult Stem Cells



DNA Methylation Differentiates Totipotent Embryonic Stem Cells from Unipotent Adult Stem Cells

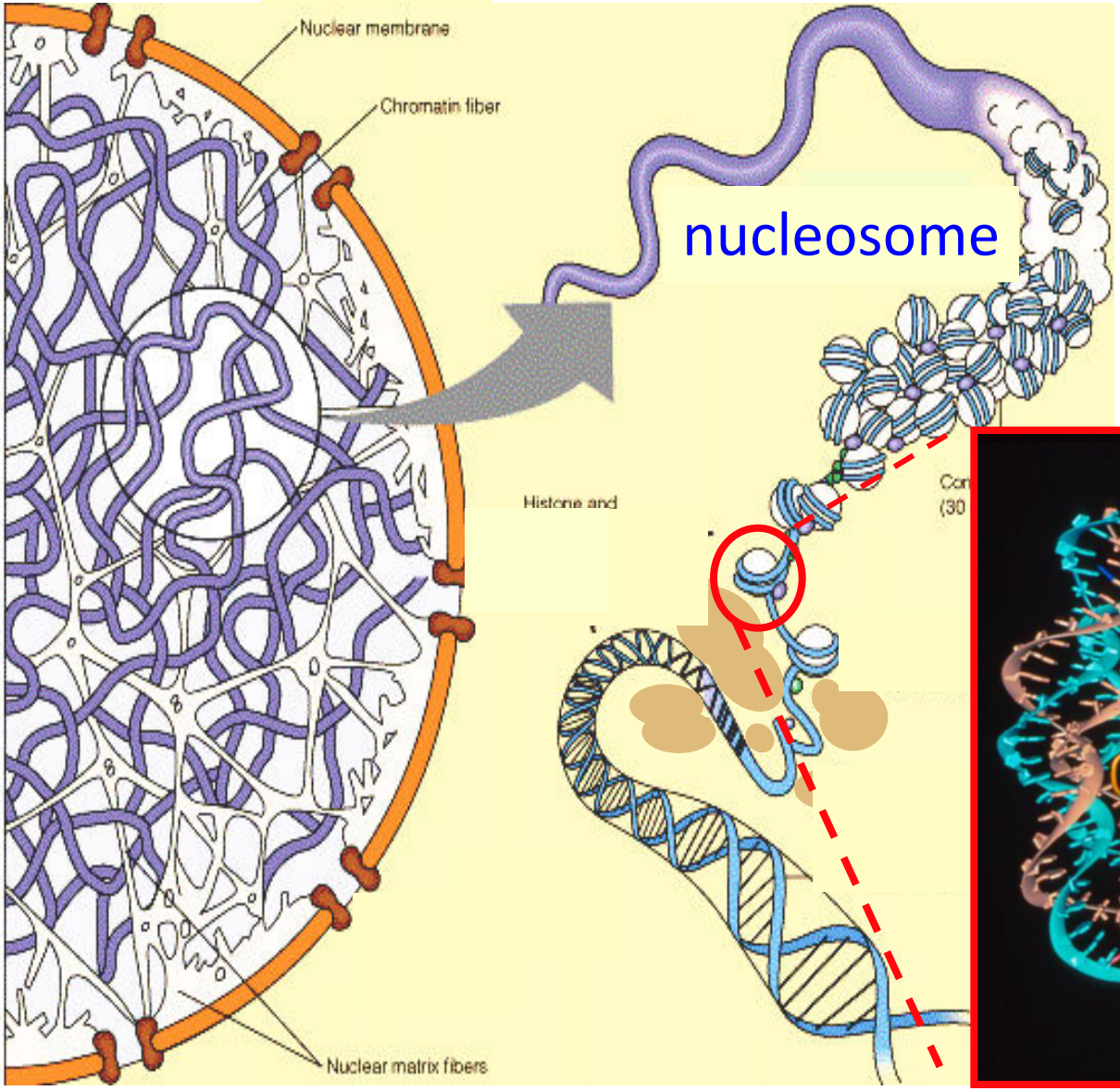


DNA Methylation Differentiates Totipotent Embryonic Stem Cells from Unipotent Adult Stem Cells



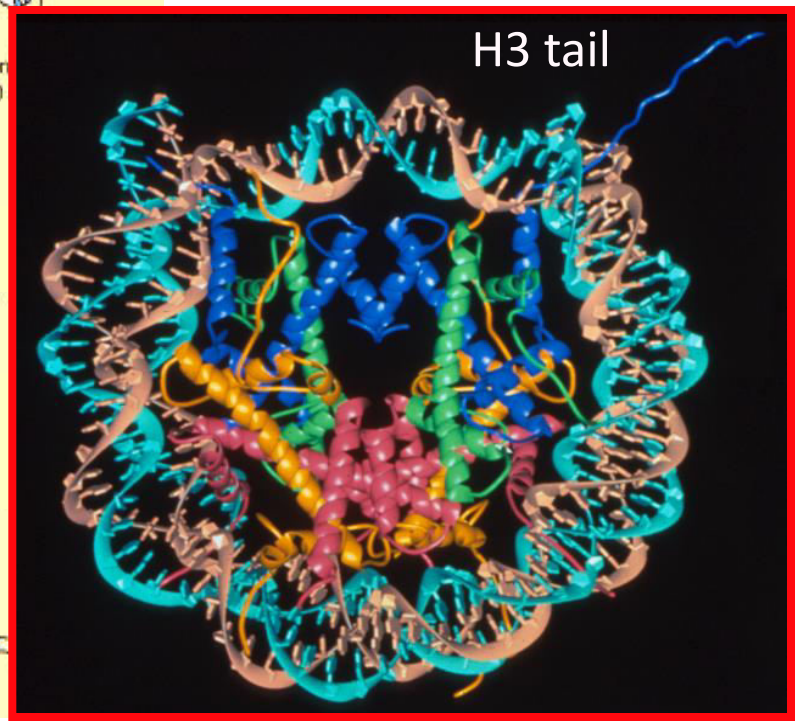
Histone modification

Chromatin structure



H2A H3

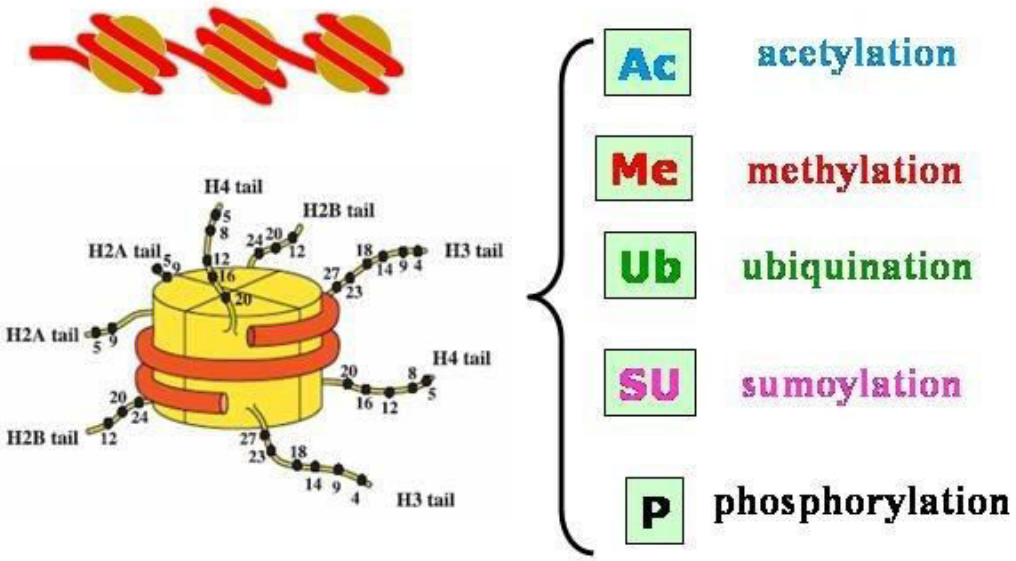
H2B H4



Luger, Richmond & colleagues 1997

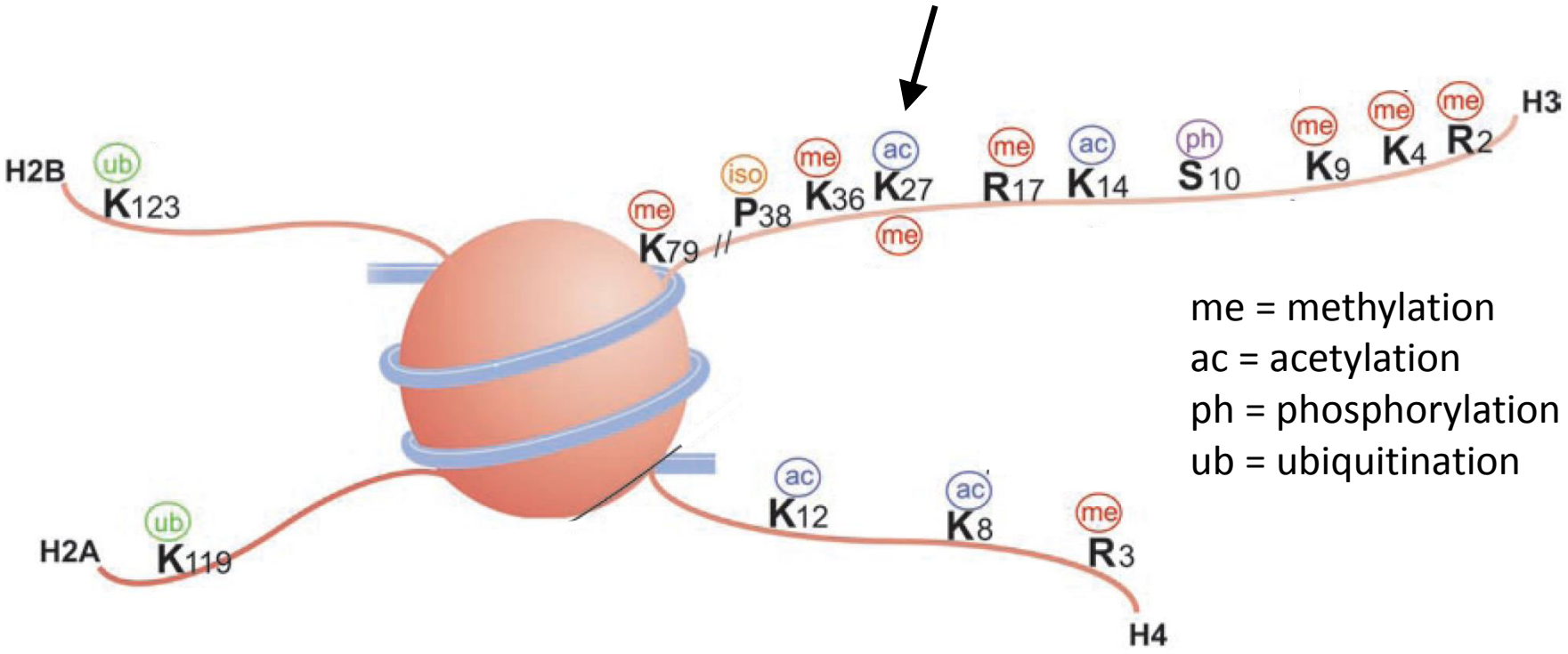
Histone modification

N-termini (tails) are particularly highly modified



The figure illustrates nucleosome models and major posttranslational modifications which play essential roles in gene expression regulation and disease processes

Covalent modifications on histones



H3 Histone

K27 Residue

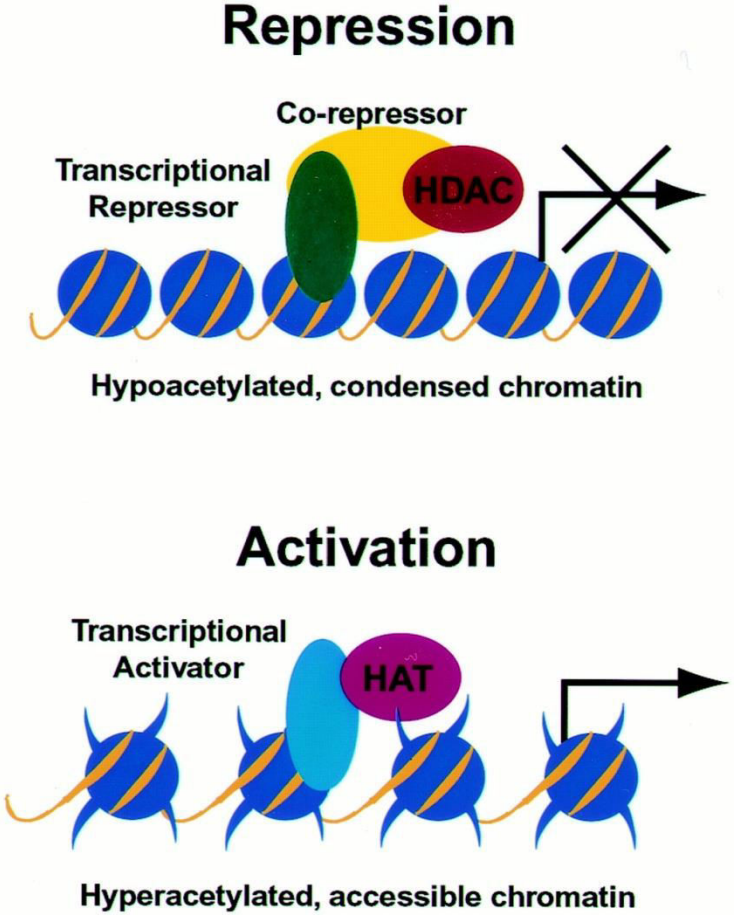
Ac Modification

From Bannister and Kouzarides (2011) Cell Res 21:381

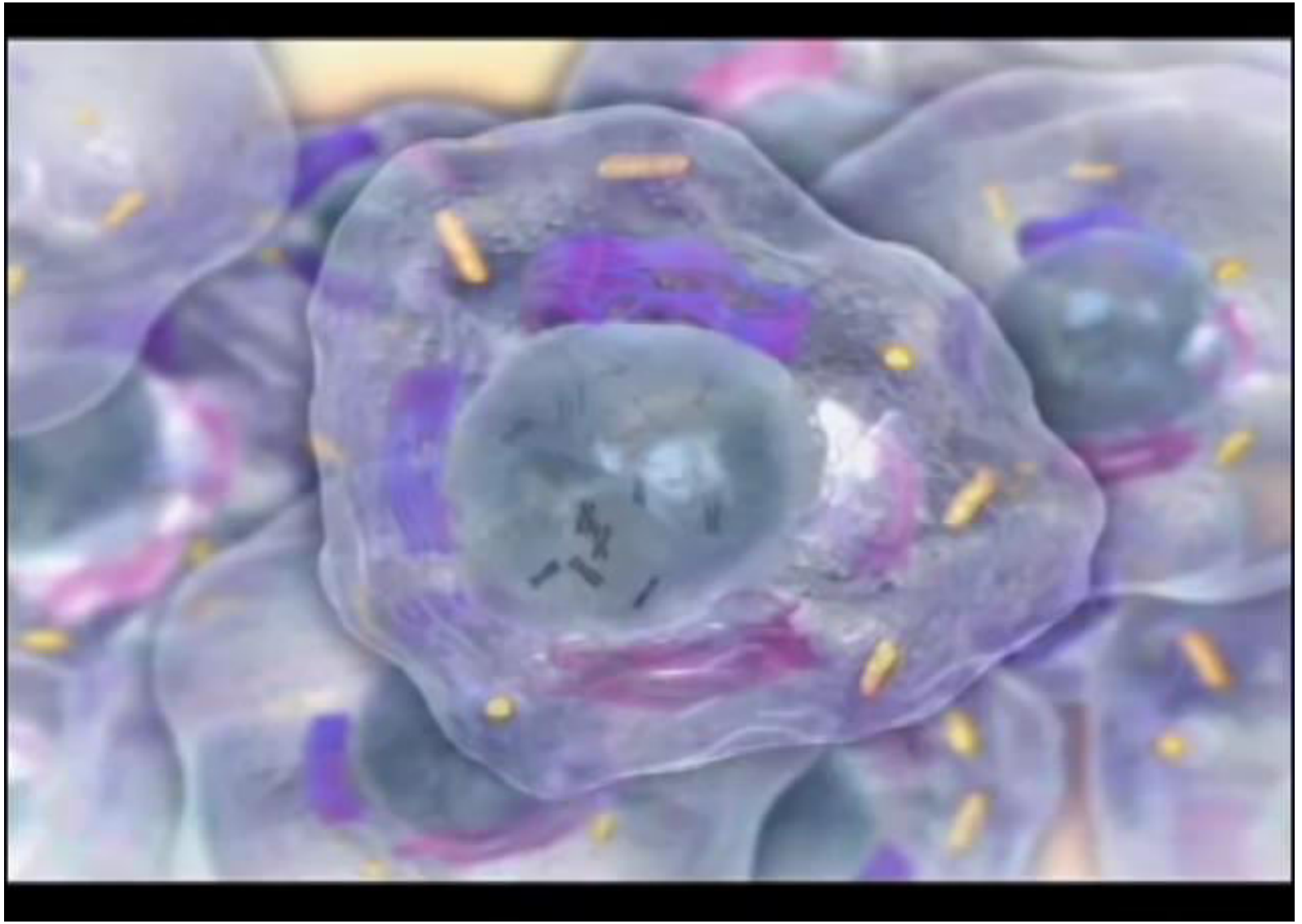
Histone Modifications and Modifers

Residue	Modification	Modiying Enzyme
Lysine	Acetylation Deacetylation	HAT HDAC
Lysine	Methylation Demethylation	HMT HDM
Lysine	Ubiquitylation Deubiquitylation	Ub ligase Ub protease
Serine/Threonine	Phosphorylation Dephosphorylation	Kinase Phosphatase
Arginine	Methylation Demethylation	PRMT Deiminase/Demethylase

Roles of Histone Modifications In Gene regulation



Watch clip



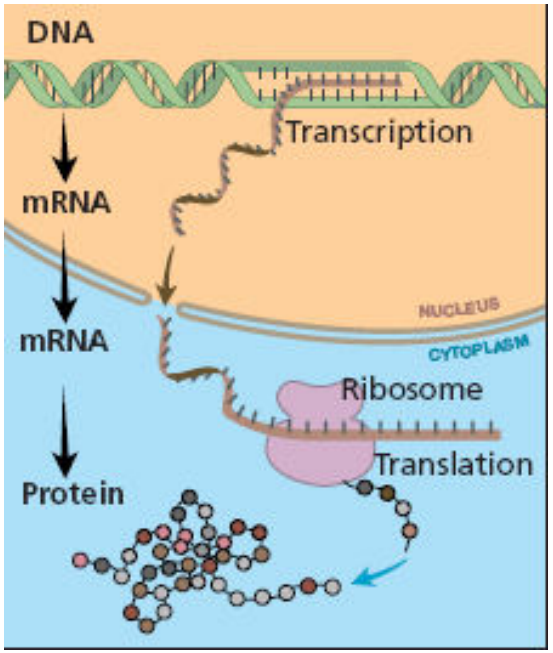
Interference RNA

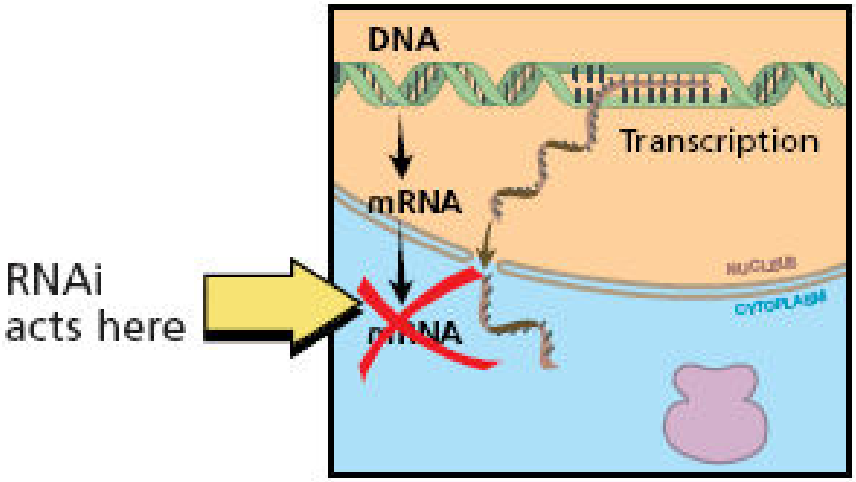
Definition

- RNA interference (RNAi) is a mechanism that inhibits gene expression at the stage of translation or by hindering the transcription of specific genes.

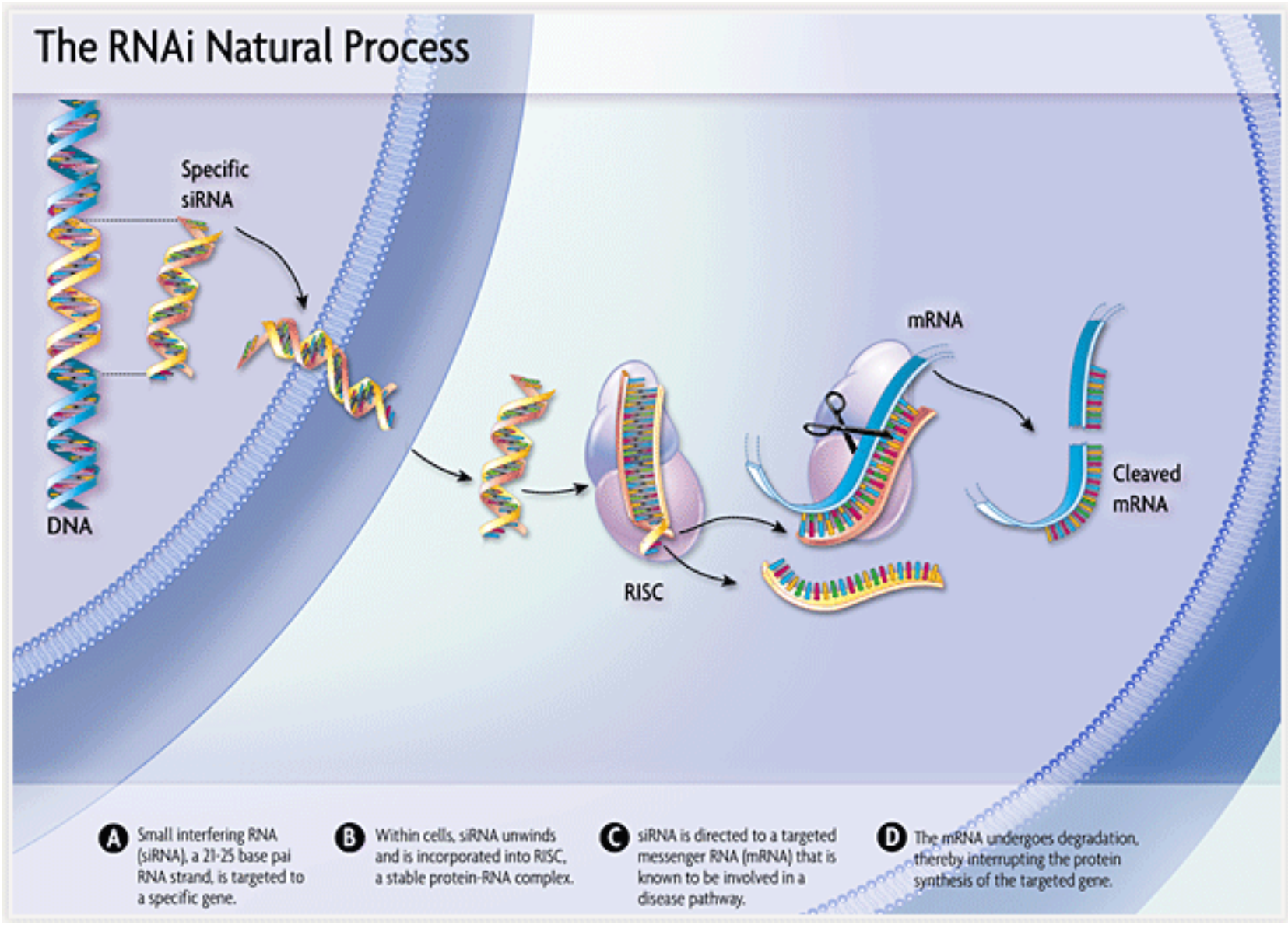
What is RNA interference (RNAi)?

- *"The Process by which dsRNA silences gene expression..."* Degradation of mRNA or translation inhibit·





In RNA interference, RNA in double-stranded form breaks down the mRNA for a specific gene, thus stopping production of protein.


The RNAi Natural Process
Specific siRNA
DNA
RISC
mRNA
Cleaved mRNA
A Small interfering RNA (siRNA), a 21-25 base pai RNA strand, is targeted to a specific gene.
B Within cells, siRNA unwinds and is incorporated into RISC, a stable protein-RNA complex.
C siRNA is directed to a targeted messenger RNA (mRNA) that is known to be involved in a disease pathway.
D The mRNA undergoes degradation, thereby interrupting the protein synthesis of the targeted gene.

Watch clip